DATA MINING AND MACHINE LEARNING ASSIGNMENT

Applying Machine Learning Models on TCGA-BRCA RNA-Seq Data
(Clustering, Classification, and Evaluation)

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1. Introduction

This report extends the TCGA-BRCA RNA-Seq data-mining pipeline. Using the cleaned, transformed, and dimensionality-reduced data three machine learning tasks are performed: (1) unsupervised clustering to discover natural groupings without using tissue-type labels; (2) supervised classification to predict tumor vs. normal tissue type; and (3) model evaluation using held-out test data, confusion matrices, and ROC-AUC curves. All 878 samples and their sample_group labels (775 Tumor, 100 Normal, 3 Metastatic) are inherited directly from the preprocessing pipeline.

2. Unsupervised Clustering

Clustering is applied without using the sample_group labels. The goal is to assess whether the data's own geometric structure in PC space recovers the known biological groupings.

2.1 K-Means Clustering

K-Means partitions samples into k clusters by iteratively assigning each sample to its nearest centroid and re-computing centroids until convergence. The algorithm minimizes total within-cluster sum of squares (inertia). Two criteria are used jointly to select k:

Elbow plot: inertia is plotted against k = 2 to 8. The optimal k is where the rate of decrease bends sharply (the elbow).

Silhouette score: measures cohesion (mean distance to own cluster) vs. separation (mean distance to nearest other cluster) for each sample. Range −1 to +1; higher is better. The k with the highest mean silhouette is selected.

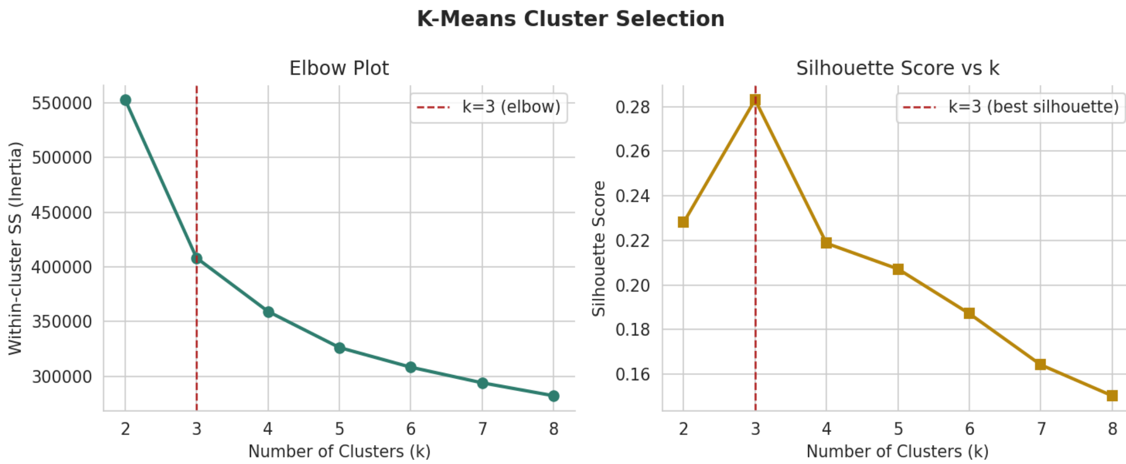

Figure 1. Elbow plot (left) and silhouette score by k (right). Both criteria converge on k = 3 as optimal

Both criteria agree: the elbow inflects at k = 3 and the silhouette score peaks at k = 3 (score = 0.283). This aligns with the three known tissue-type groups, suggesting the expression data's geometric structure reflects the underlying biology.

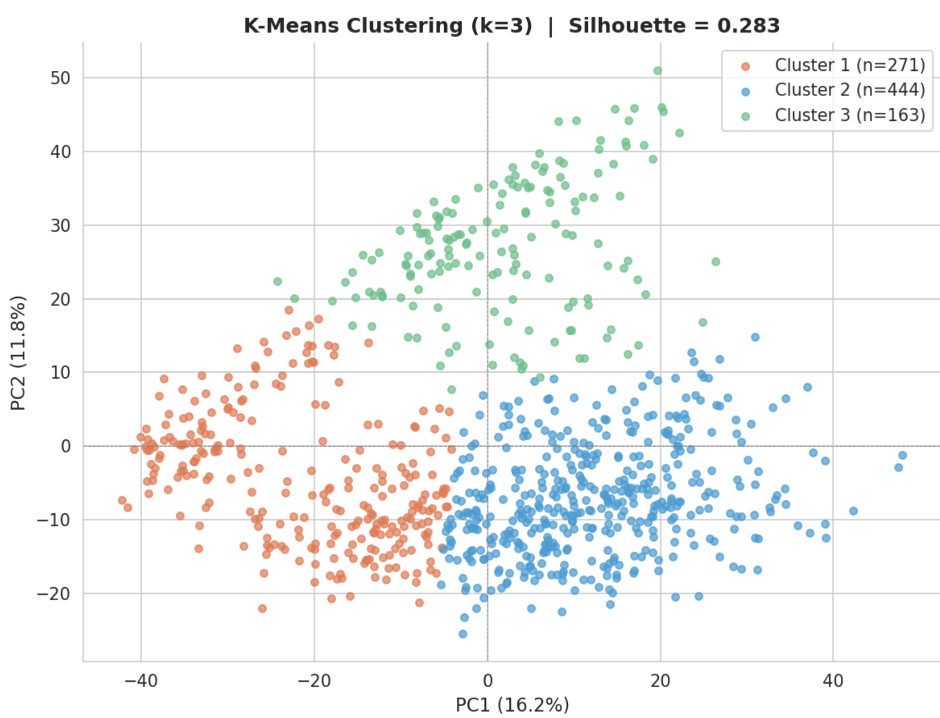

Figure 2. K-Means (k=3) cluster assignments projected onto PC1-PC2. Cluster 2 captures the majority of tumor samples on the positive PC1 side; Cluster 1 captures most normal samples on the negative PC1 side

Cross-tabulating the discovered clusters against the known tissue groups (Table 1) confirms the alignment. Cluster 2 is almost exclusively tumor (441 Tumor, 0 Normal), while Cluster 1 contains 96 of the 100 Normal samples together with 175 Tumor samples that overlap in PC space with normal tissue, consistent with these being the more normal-like Luminal A tumors. All 3 Metastatic samples fall in Cluster 2, consistent with their tumor-tissue origin.

Table 1. Cross-tabulation of K-Means cluster assignments against known tissue-type groups

K-Means Cluster	Metastatic	Normal	Tumor
Cluster 1	0	96	175
Cluster 2	3	0	441
Cluster 3	0	4	159

2.2 Hierarchical Clustering

Hierarchical clustering with Ward's linkage builds a tree (dendrogram) of nested groupings without requiring k in advance. Ward's criterion merges the pair of clusters that produces the smallest increase in total within-cluster variance which is the same objective K-Means minimizes, making the two methods directly comparable. The dendrogram is cut at the height that yields k = 3 clusters.

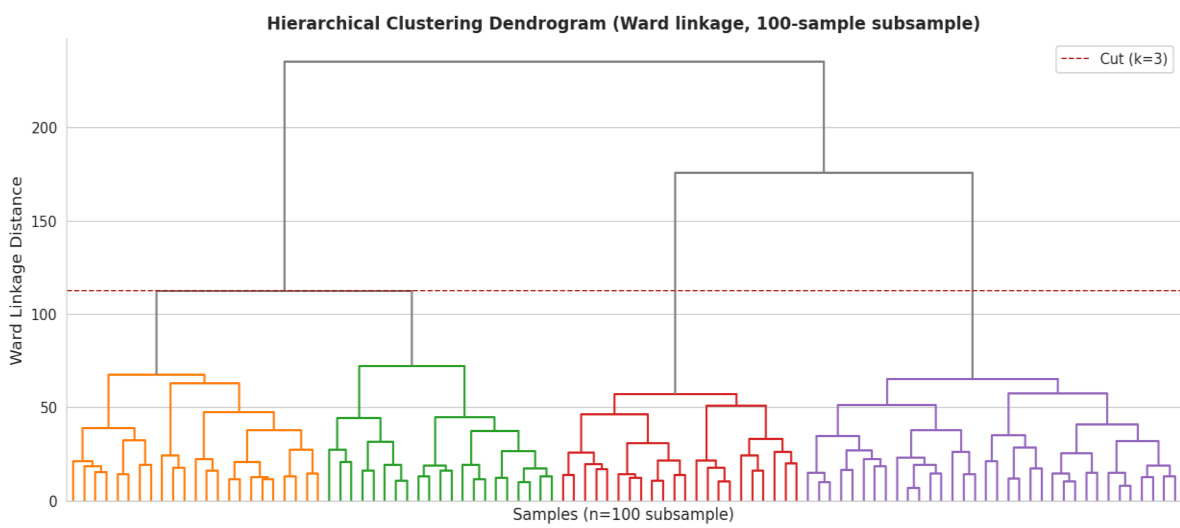

Figure 3. Dendrogram from hierarchical clustering (Ward linkage) on a 100-sample subsample. The red dashed line marks the cut height producing k = 3 clusters

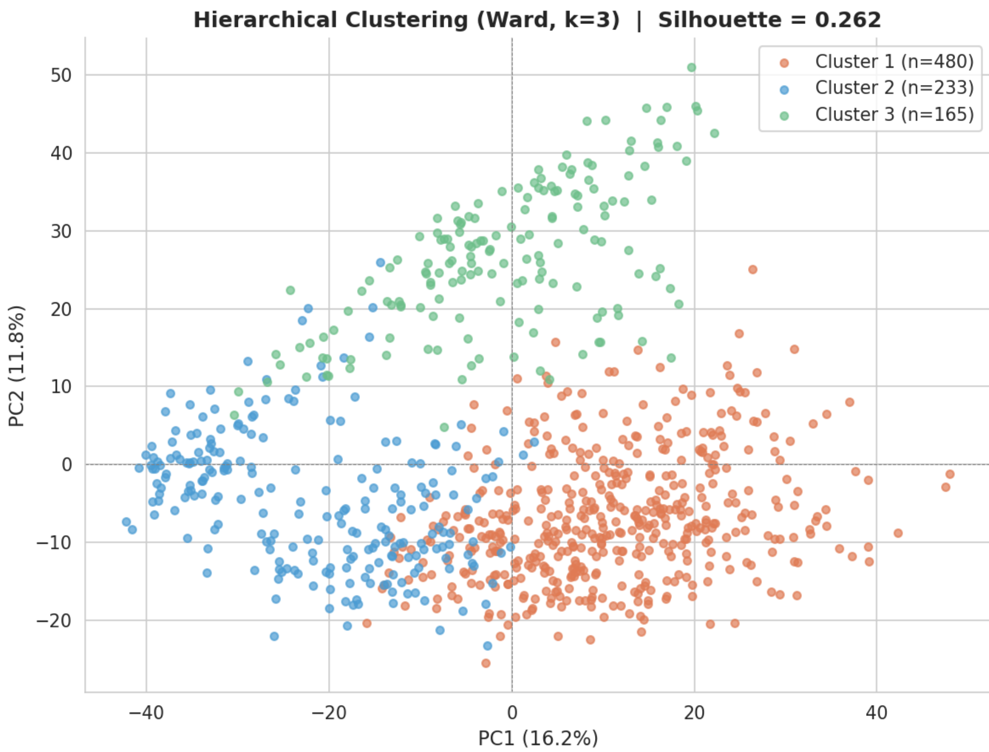

Figure 4. Hierarchical clustering (Ward, k=3) projected onto PC1-PC2. The cluster structure closely mirrors the K-Means result

The hierarchical solution (silhouette = 0.262) closely mirrors the K-Means result (silhouette = 0.283), indicating that the three-cluster structure is stable across algorithms. The moderate silhouette values for both methods reflect the genuine within-tumor heterogeneity arising from breast cancer molecular subtypes. The clusters capture the dominant biological structure but cannot fully separate the overlapping tumor subgroups visible along PC2.

Table 2. Clustering performance summary

Method	k	Silhouette Score	Interpretation
K-Means	3	0.283	Good separation; clusters stable across random starts
Hierarchical (Ward)	3	0.262	Consistent with K-Means; confirms 3-cluster structure

3. Supervised Classification

Three classifiers are trained to predict tissue type (Tumor vs. Normal) from the first 10 PC scores. The 3 Metastatic samples are excluded: with only 3 observations they cannot form a reliable held-out evaluation set, and their inclusion would create severe class imbalance that distorts precision and recall estimates.

3.1 Train/Test Split

An 80/20 stratified random split is applied, yielding 700 training samples (620 Tumor, 80 Normal) and 175 test samples (155 Tumor, 20 Normal). Stratification preserves the approximately 88:12 tumor-to-normal ratio in both subsets, ensuring the minority class is proportionally represented in the evaluation set.

3.2 Logistic Regression (Ridge-penalized)

Logistic Regression models the log-odds of tumor membership as a linear combination of the 10 PC features. Ridge (L2) regularization shrinks coefficients toward zero, reducing variance and preventing overfitting, particularly valuable here given the class imbalance and the small Normal class. Five-fold cross-validation is used on the training set to tune regularization. Logistic Regression serves as the interpretable baseline: its PC coefficients directly quantify each component;s contribution to the tumor/normal decision boundary.

3.3 Random Forest

Random Forest builds an ensemble of 500 decision trees, each trained on a bootstrapped sample of the training data with a random subset of the 10 PCs considered at each split. Predictions are made by majority vote across trees. The ensemble approach reduces variance (overfitting) without increasing bias, and the out-of-bag samples provide built-in training-set validation. Feature importance (measured as mean decrease in Gini impurity) quantifies each PC's contribution to classification (Figure 5).

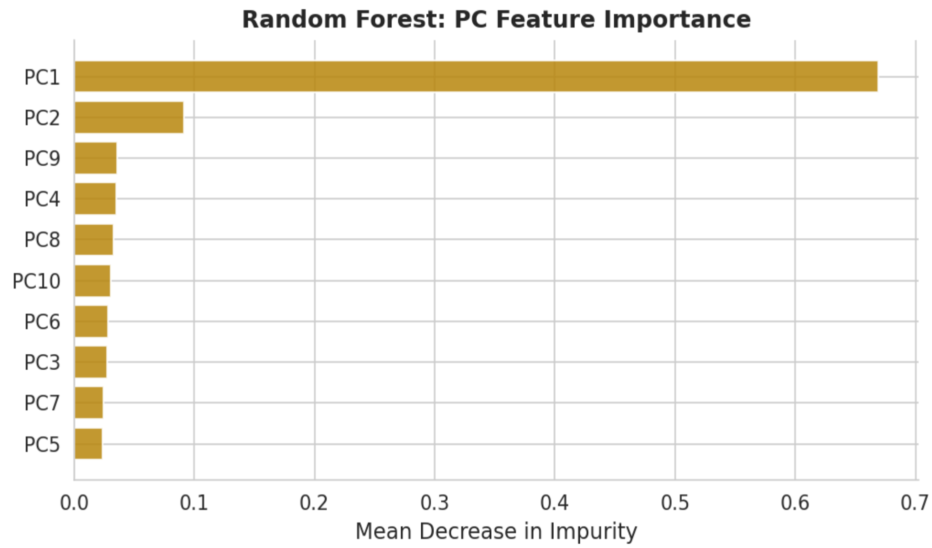

Figure 5. Random Forest feature importance (mean decrease in Gini impurity). PC1, which captures the dominant tumor/normal transcriptomic axis, is the most discriminative feature

3.4 Support Vector Machine (RBF Kernel)

SVM finds the maximum-margin hyperplane separating Tumor from Normal in the 10-dimensional PC space. The Radial Basis Function (RBF) kernel implicitly maps samples into a higher-dimensional space, enabling non-linear decision boundaries that linear classifiers cannot form. Parameters C = 10 (controls the margin-violation penalty) and gamma = scale (= 1 / (n_features × variance of X)) balance bias and variance. In the context of this dataset, SVM's kernel trick is motivated by the within-tumor subtype structure that creates non-linear boundaries in PC space.

4. Model Evaluation

4.1 Confusion Matrices

Confusion matrices display the four classification outcomes on the held-out test set: True Positives (TP: tumor correctly identified), True Negatives (TN: normal correctly identified), False Positives (FP: normal misclassified as tumor), and False Negatives (FN: tumor missed as normal). In a cancer detection context, minimizing FN (missed tumors) is clinically paramount.

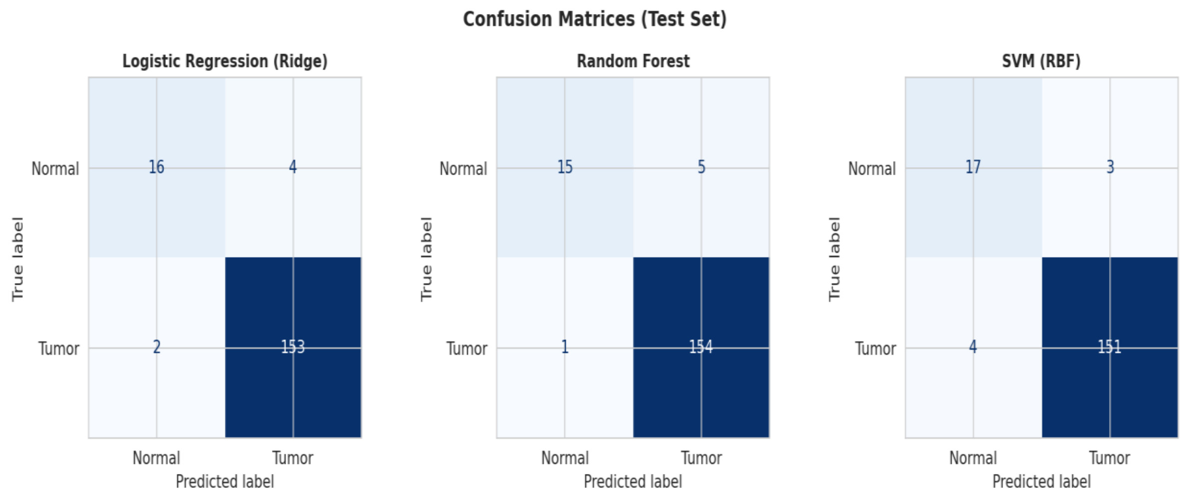

Figure 6. Confusion matrices for all three classifiers on the 175-sample test set

All three classifiers show very low false-negative counts (2-4 missed tumors out of 155). Random Forest achieves the lowest FN count (1 missed tumor), making it the safest choice for a clinical screening application. False positives (normal classified as tumor: 3-5) are slightly higher across all models, reflecting the challenge of the imbalanced class boundary.

4.2 ROC Curves

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (sensitivity) against the False Positive Rate (1 − specificity) across all probability decision thresholds. The Area Under the Curve (AUC) summarises discrimination in a single threshold-independent metric: AUC = 1.0 is perfect, AUC = 0.5 is random. AUC is particularly informative for imbalanced datasets because it evaluates the full probability ranking, not just the default 0.5 threshold.

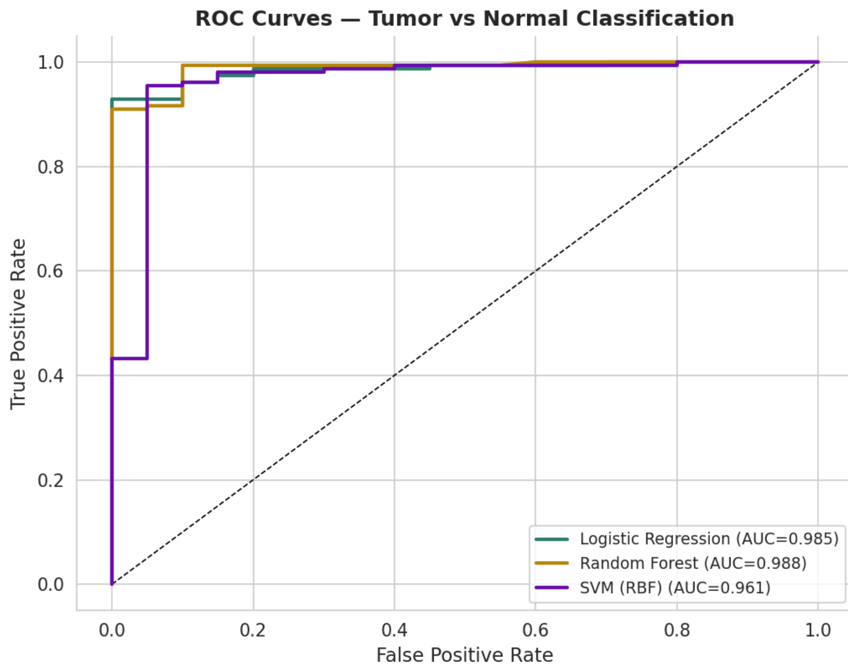

Figure 7. ROC curves for all three classifiers. All models achieve AUC > 0.96, indicating excellent discrimination between Tumor and Normal samples

4.3 Performance Summary

Five evaluation metrics are reported for each classifier on the held-out test set:

Accuracy: proportion of all predictions correct.

Precision: of all samples predicted as Tumor, what fraction truly are.

Recall (Sensitivity): of all true Tumors, what fraction are correctly identified.

F1 score: harmonic mean of Precision and Recall, balancing the two.

ROC-AUC: threshold-independent discrimination metric.

Table 3. Classification performance on the held-out test set

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression (Ridge)	0.966	0.975	0.987	0.981	0.985
Random Forest	0.966	0.969	0.994	0.981	0.988
SVM (RBF)	0.960	0.981	0.974	0.977	0.961

6. Conclusion

Both clustering methods converged on k = 3 as the natural grouping of TCGA-BRCA samples in the 10-dimensional PC space, consistent with the three known tissue-type categories. The moderate silhouette scores (K-Means: 0.283; Hierarchical: 0.262) reflect genuine within-tumor heterogeneity from breast cancer molecular subtypes; the clusters recover the dominant biological structure but cannot fully resolve the overlapping subgroups visible along PC2. The close agreement between K-Means and Ward hierarchical clustering confirms the result is not algorithm-dependent.

All three classifiers achieved high accuracy (≥96.0%) and ROC-AUC (≥0.961) on the held-out test set, confirming that the first 10 PCs contain sufficient discriminative information to distinguish Tumor from Normal tissue reliably. Random Forest achieved the highest AUC (0.988) and recall (0.994), making it the strongest overall classifier for this task, particularly important in a cancer detection context where missing a true tumor is more costly than a false alarm. Logistic Regression matched Random Forest on accuracy and F1 while providing full interpretability through its PC coefficients. SVM was competitive on precision (0.981) and robust across all metrics.

The consistently high performance across three structurally different classifiers provides strong evidence that the transcriptomic signal preserved through the preprocessing, transformation, and PCA pipeline is not only biologically meaningful but also machine-learning exploitable. These results support the use of this pipeline as a foundation for more complex downstream tasks, such as molecular subtype classification or survival-risk prediction, where the richer subtype structure within the tumor group could be explicitly targeted.

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